

Chapter 01 – Inception

Theory – Answers

1. What is Emmet?

-> Emmet is the essential toolkit for web-developers.

It allows you to type shortcuts that are then expanded into

Full-fledged boiler plate code for writing HTML and CSS.

2. Difference between a Library and Framework?

-> 1. "A library is a collection of reusable functions that we call when needed, while a framework provides a complete structure and controls the application flow."

2. A key difference between the two is Inversion of Control.

When using a library, the control remains with the developer who tells the

Application when to call library functions. When using a framework,

The control is reversed, which means that the framework tells the

Developer where code needs to be provided and calls it as it requires.

Library

You call the library when needed

Gives specific functionality

More freedom

Example: React, jQuery

Framework

Framework calls your code

Gives full structure to the app

More rules/conventions

Example: Angular, Django

3. What is CDN? Why do we use it?

-> CDN stands for Content Delivery Network

It is a network of servers distributed across different locations

That delivers files like: JavaScript, CSS, Images, Videos, React Libraries

Faster to users.

In simple words....

"CDN (Content Delivery Network) is a distributed network of servers used to deliver files like JavaScript, CSS, images, and videos faster by serving them from the nearest server location."

4. Why is React known as React?

-> React is a JavaScript Library. The name 'React' was chosen

Because the library was designed to allow developers to react

To changes in state and data within an application, and to

Update the user interface in a declarative and efficient manner.

5. What is crossorigin in script tag?

-> The crossorigin attribute in the script tag enables

Cross-Origin Resource Sharing (CORS) for loading

External JavaScript files from different origin than the hosting

Web page. This allows the script to access resources from

The server hosting the script, such as making HTTP requests or accessing data.

6. What is difference between React and ReactDOM?

->1. "React is used to build UI components, while ReactDOM is used

To render those components into the browser DOM.

2. React :- Creates elements/Components

Eg. Function App() {

 Return<h1> Hello </h1>;

}

3. ReactDOM :- Takes those components and display them inside

The browser.

Eg. ReactDOM.createRoot(document.getElementById("root")).render(<App />);

React

React is a JavaScript library for building UI

Used to create components

Handles UI logic

Package: react

ReactDOM

ReactDOM connects React with the browser DOM

Used to render components on the webpage

Handles DOM rendering

Package: react-dom

7. What is difference between react.development.js and react.production.js files via CDN?

-> react.development.js :- Used while learning/building.

It provides :- Detailed errors, warnings, debugging help

react.production.js :- Used when deploying the app.

It is minified, optimized, faster, no extra warnings/debug info

In short we can explain like this:

"react.development.js is used during development because it provides warnings and debugging information, while react.production.js is optimized and minified for better performance in production."

react.development.js

Used during development

Includes warnings and error messages

Easier debugging

Larger file size

Slower

Human-readable code

react.production.js

Used in production/live apps

Optimized and compressed

Faster performance

Smaller file size

Faster

Minified code

8. What is async and defer attributes?

->1. "async loads a script in parallel and executes it immediately after downloading, while defer also loads in parallel but executes only after the HTML document is fully parsed.

2. Async and defer are attributes used in the <script> tag to load JavaScript files

More efficiently without blocking HTML parsing.

3. Problem without async/ defer

Normal script: <script src="app.js"></script>

Browser behavior: 1. Stops HTML parsing

2. Downloads JS

3. Executes JS

4. Continues HTML parsing.

This can slow down page loading.

4. async : <script async src="app.js"></script>

It works like; 1.Script downloads in the background

2. Execute immediately after download

3. HTML parsing may pause during execution.

Note that, Execution order is NOT guaranteed.

Best for ads, analytics, independent scripts

5. defer: <script defer src="app.js"></script>

It works like; 1. Script downloads in the background.

2. Executes only after HTML parsing is complete.

Note that, Execution order IS maintained.

Best for main application script, React projects, scripts depending on DOM

Important Tip

For multiple scripts that depends on each other

- Use defer
- Avoid async , because async execution order can change

async

Downloads in background

Executes immediately after download

May interrupt HTML parsing

Order not guaranteed

Good for independent scripts

defer

Downloads in background

Executes after HTML parsing

Does not block parsing

Order maintained

Good for app scripts